

Micro aflevering 2

a) Opstil problemet

$$u(x_1, x_2) = x_1 + 2x_1x_2 + x_2$$

$$s.t. \quad p_1x_1 + p_2x_2 = M$$

b) Brug lagrange metoden til at finde x_1^* og x_2^*

$$L(\lambda, x_1, x_2) = u(x_1, x_2) - \lambda(p_1x_1 + p_2x_2 - M)$$

$$(1) \quad \frac{\partial L}{\partial x_1} = 1 + 2x_2 - \lambda p_1 = 0$$

$$(2) \quad \frac{\partial L}{\partial x_2} = 1 + 2x_1 - \lambda p_2 = 0$$

$$(3) \quad \frac{\partial L}{\partial \lambda} = p_1x_1 + p_2x_2 - M = 0$$

Fra 1 og 2 har vi

$$\frac{1 + 2x_2}{1 + 2x_1} = \frac{p_1}{p_2}$$

$$1 + 2x_2 = \frac{p_1}{p_2} * (1 + 2x_1)$$

$$x_2 = \frac{p_1}{p_2} \left(\frac{1}{2} + x_1 \right) - \frac{1}{2}$$

Fyldes ind i 3 som løses for x_1

$$p_1x_1 + p_2x_2 = M \leftrightarrow p_1x_1 + p_2 * \left(\frac{p_1}{p_2} \left(\frac{1}{2} + x_1 \right) - \frac{1}{2} \right) = p_1x_1 + \frac{1}{2}p_1 + p_1x_1 - \frac{1}{2}p_2 = M$$

$$2p_1x_1 = M - \frac{1}{2}p_1 + \frac{1}{2}p_2$$

$$x_1^* = \frac{M}{p_1 * 2} - \frac{1}{4} + \frac{1}{4} \frac{p_2}{p_1}$$

$$x_2^* = \frac{p_1}{p_2} \left(\frac{1}{2} + x_1^* \right) - \frac{1}{2} = \frac{p_1}{p_2} \left(\frac{1}{2} + \frac{M}{p_1 * 2} - \frac{1}{4} + \frac{1}{4} \frac{p_2}{p_1} \right) - \frac{1}{2} = \frac{1}{4} \frac{p_1}{p_2} + \frac{M}{p_2 * 2} - \frac{1}{4}$$

C+D) Find den indre nyttefunktion

$$\begin{aligned} U(x_1^*, x_2^*) &= \frac{M}{p_1 * 2} - \frac{1}{4} + \frac{1}{4} \frac{p_2}{p_1} + \frac{1}{4} \frac{p_1}{p_2} + \frac{M}{p_2 * 2} - \frac{1}{4} + 2 * \left(\frac{1}{4} \frac{p_1}{p_2} + \frac{M}{p_2 * 2} - \frac{1}{4} \right) * \left(\frac{M}{p_1 * 2} - \frac{1}{4} + \frac{1}{4} \frac{p_2}{p_1} \right) \\ &= \left(\frac{m^2}{2 * p_1 * p_2} \right) + m * \left(\left(\frac{1}{2 * p_1} \right) + \left(\frac{1}{2 * p_2} \right) \right) + \left(\frac{p_1}{8 * p_2} \right) + \left(\frac{p_2}{8 * p_1} \right) - \left(\frac{1}{4} \right) = V(p_1, p_2, M) \end{aligned}$$

Som er en funktion af indkomsten M og priserne p_1 og p_2

e) er x_1^* og x_2^* normale goder
tjekker definitionen

$$x_1^* = \frac{M}{p_1 * 2} - \frac{1}{4} + \frac{1}{4} \frac{p_2}{p_1}$$

$$\frac{dx_1^*}{dM} = \frac{1}{p_1 * 2} \geq 0$$

$$x_2^* = \frac{1}{4} \frac{p_1}{p_2} + \frac{M}{p_2 * 2} - \frac{1}{4}$$

$$\frac{dx_2^*}{dM} = \frac{1}{p_2 * 2} \geq 0$$

De er begge normale goder

f) er x_1^* og x_2^* ordinære goder
tjekker definitionen

$$x_1^* = \frac{M}{p_1 * 2} - \frac{1}{4} + \frac{1}{4} \frac{p_2}{p_1}$$

$$\frac{dx_1^*}{dp_1} = -\frac{M}{2} \left(\frac{1}{p_1^2} \right) - \frac{p_2}{4} \left(\frac{1}{p_1^2} \right) \leq 0$$

$$x_2^* = \frac{1}{4} \frac{p_1}{p_2} + \frac{M}{p_2 * 2} - \frac{1}{4}$$

$$\frac{dx_2^*}{dp_2} = -\frac{M}{2} \left(\frac{1}{p_2^2} \right) - \frac{p_1}{4} \left(\frac{1}{p_2^2} \right) \leq 0$$

g) er x_1^* *kompliment eller substitut* for x_2^*
tjek definitionen

$$x_1^* = \frac{M}{p_1 * 2} - \frac{1}{4} + \frac{1}{4} \frac{p_2}{p_1}$$

$$\frac{dx_1^*}{dp_2} = \frac{1}{4 * p_1} \geq 0$$

altså må x_1 være substitut for x_2 og vice versa da det kan gøres analogt